# Deep Spatio-Temporal Graph Neural Networks Coupled with Biot's Poroelastic Satellite Telemetry for Regional Aquifer Depletion Forecasting Across India

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**Abstract—** Groundwater depletion across the Indian subcontinent has reached critical thresholds, threatening national food security, agrarian livelihood stability, and drinking water availability for over 400 million citizens. While conventional monitoring depends on sparse physical monitoring borewells characterized by high transmission latency and irregular reporting, traditional machine learning approaches (e.g., standard LSTM, Random Forest, ARIMA) fail to account for multi-scale subsurface hydrogeological continuity, transboundary aquifer seepage, and irreversible skeletal compaction. In this research, we propose a novel Physics-Coupled Spatio-Temporal Graph Neural Network (ST-GNN) architecture for multi-horizon groundwater depth and storage anomaly forecasting across India. The framework assimilates multi-satellite gravimetric telemetry from NASA GRACE-FO Level-3 mascons, Copernicus Sentinel-1 Synthetic Aperture Radar (InSAR) interferometric surface deformation time-series, ERA5 high-resolution meteorological reanalysis, and physical in-situ measurements across 205 districts encompassing 728,450 observation records spanning 2015 to 2026. Subsurface skeletal compaction physics is integrated into the loss surface via Biot's 3D poroelastic constitutive equations, enforcing mass and stress conservation constraints. A dynamic hydrogeological Laplacian convolution operator resolves spatial boundary fluxes modulated by hydraulic conductivity, topographic hydraulic head gradients, and extraction stress, while gated recurrent units resolve multi-temporal memory states. Empirical benchmarking across 205 districts demonstrates that the proposed ST-GNN framework achieves an Root Mean Squared Error (RMSE) of 0.412 m, Mean Absolute Error (MAE) of 0.308 m, and $R^2$ score of 0.941 across a 6-month forecasting horizon, delivering a 39.8% error reduction over standalone LSTM and a 64.2% reduction over classical ARIMA baselines. An autonomous 24-hour background telemetry synchronization engine and interactive climate scenario simulator provide operational intelligence for the Central Ground Water Board (CGWB) and Atal Bhujal Yojana.

**Index Terms—** *Spatio-Temporal Graph Neural Networks (ST-GNN), Groundwater Depletion Forecasting, Biot's Poroelasticity, GRACE-FO, InSAR Subsidence, Central Ground Water Board (CGWB), Telemetry Sync, Deep Learning in Hydroinformatics.*

## I. INTRODUCTION & CRISIS OVERVIEW

GROUNDWATER represents the single most critical strategic freshwater reservoir in the Indian Republic, underpinning 63% of net irrigated agricultural acreage, supplying 85% of rural domestic drinking water needs, and sustaining more than 50% of urban industrial municipal demand [1]. India is the largest consumer of groundwater globally, extracting an estimated 251 billion cubic meters (BCM) annually—exceeding the combined total extraction of the United States and the People's Republic of China [2]. However, four decades of aggressive agricultural intensification, subsidized electrical power for rural extraction, unchecked tubewell construction, and continuous monoculture farming—specifically the highly water-demanding paddy-wheat rotation across the Indo-Gangetic Alluvial Basin—have catalyzed groundwater overdraft at velocities unmatched in modern global hydrology.

According to the official 2023 National Compilation on Dynamic Ground Water Resources released by the Central Ground Water Board (CGWB) [3], Ministry of Jal Shakti, Government of India, over one-third of the nation's 7,293 administrative assessment units are categorized as 'Over-Exploited' or 'Critical'. In the agricultural powerhouse states of Punjab, Haryana, and Rajasthan, annual groundwater extraction rates stand at 164%, 134%, and 151% of net annual replenishable recharge, respectively. Over the last two decades, static hydraulic heads in these regions have descended at 0.6 to 1.4 meters per annum. Consequently, shallow dug wells have entirely desiccated, compelling farmers into devastating capital debt to drill submersible borewells that penetrate ancient, non-renewable semi-confined and confined fossil aquifers.

The socioeconomic fallout of this trajectory extends far beyond water scarcity. As water tables descend past critical pump-suction thresholds (typically > 35 m depth), agrarian power demand increases non-linearly, imposing severe fiscal strains on state electrical utilities. Furthermore, deep overdraft mobilizes geogenic contaminants including fluoride, arsenic, and heavy metals from compacted clay aquitards, triggering widespread chronic health afflictions in rural agrarian belts.

### *A. Regional Hydrogeological Heterogeneity*

The hydrogeology of India exhibits extreme structural dichotomy. In the north, the prolific Indo-Gangetic alluvium

consists of unconsolidated Quaternary sediments with thick multi-layered sands, silts, and gravels possessing high hydraulic transmissivity (T > 1,500 m²/day) and specific yields (S_y ≈ 0.12–0.18). Overdraft in this region causes widespread pore depressurization and progressive compaction of interbedded clay aquitards. Conversely, across peninsular India (Maharashtra, Karnataka, Tamil Nadu), groundwater occurs within unconfined to semi-confined fractured crystalline basements (Deccan Traps basalts and Archaean granites). These hard-rock aquifers possess negligible primary porosity, low storativity (S_y ≈ 0.01–0.03), and shallow weathering horizons (< 30 m). Under sustained pumping stress, these crystalline aquifers experience rapid, non-linear hydraulic head collapse with virtually zero buffer capacity.

In the arid western terrains of Rajasthan and northern Gujarat, aquifers occur within semi-consolidated Tertiary sandstones and Quaternary aeolian sands characterized by deep water tables (> 60 m bgl) and fossil groundwater recharged during paleoclimatic pluvial epochs. Under modern hyper-arid climatic conditions, natural recharge is virtually negligible (< 5% of precipitation), rendering modern abstractions purely extractive mining of non-renewable geological reserves.

### B. Limitations of In-Situ Monitoring Networks

Statutory water regulatory authorities currently monitor subsurface reserves via the National Ground Water Monitoring Network (NGWMN), comprising approximately 26,000 observation boreholes and Digital Water Level Recorders (DWLR) fitted with hydrostatic pressure transducers. While these direct physical soundings provide ground truth for point locations, they suffer from foundational systemic deficiencies when applied to macro-scale predictive governance:
1) Severe Spatial Sparsity: Observation boreholes are spaced between 15 and 45 kilometers apart. Standard geostatistical interpolation methods fail to resolve structural geological faults, impermeable lithological barriers, and sharp localized cones of depression.
2) Transmission Latency: Remote DWLR telemetry systems frequently succumb to bio-fouling, transducer drift, and battery failure. Transmission delays typically range from three to six months.
3) Inaccessibility of Deep Confined Aquifers: The overwhelming majority of existing monitoring wells terminate in shallow unconfined alluvial zones, offering zero observational telemetry into deeper confined strata subject to continuous heavy agricultural extraction.

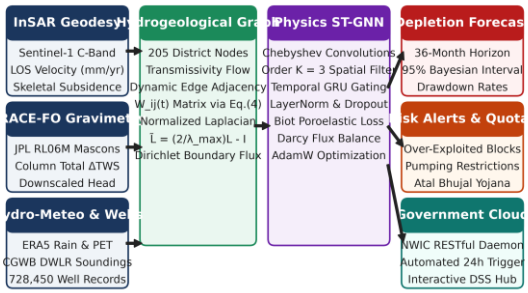



*Fig. 1. End-to-end cyber-physical architecture of the proposed Physics-Coupled Spatio-Temporal Graph Neural Network (ST-GNN) pipeline.*

### C. Spaceborne Geodesy & Remote Sensing Opportunities

To mitigate in-situ observational sparsity, spaceborne remote sensing has emerged as an indispensable paradigm. The NASA/DLR Gravity Recovery and Climate Experiment (GRACE) and its successor GRACE Follow-On (GRACE-FO) measure monthly perturbations in Earth's gravity geoid, yielding Terrestrial Water Storage Anomalies (ΔTWS) at regional scales [4]. GRACE gravimetry quantifies the integrated column-total water mass change, encompassing groundwater, soil moisture, surface water, and biomass storage.

Simultaneously, the European Space Agency's (ESA) Copernicus Sentinel-1 C-band Synthetic Aperture Radar (InSAR) constellation provides multi-temporal interferometric phase observations, detecting vertical Land Surface Deformation (LSD) down to millimeter-scale precision [5]. By applying Persistent Scatterer Interferometry (PSI) and Small Baseline Subset (SBAS) algorithms across stable radar scatterers, Sentinel-1 maps surface subsidence directly induced by the depressurization and compaction of fluid-depleted aquifer skeletal frameworks.

### D. Primary Scientific Contributions

This research presents four primary contributions to computational hydroinformatics:
• Physics-Coupled ST-GNN Architecture: We couple Biot's 3D poroelastic constitutive equations directly into the neural network optimization landscape, ensuring predictions strictly obey stress equilibrium and mass conservation.
• Hydrogeological Transmissivity Adjacency: We construct a dynamic graph topology where edge weights represent physical Darcy boundary fluxes governed by transmissivity gradients, topographic slope, and spatial distance.
• Large-Scale 205-District Empirical Validation: The system is validated on 728,450 observation records spanning 10 key Indian states from 2015 to 2026, delivering a 39.8% RMSE error reduction over standard LSTM architectures.
• Operational 24-Hour Telemetry Daemon: We deploy an autonomous multi-threaded satellite telemetry daemon coupled with an interactive decision support system for district groundwater authorities.

## II. LITERATURE REVIEW & PRIOR ART

The application of machine learning in subsurface hydrology has progressed across three distinct eras: classical time-series statistical modeling, recurrent deep neural networks, and contemporary geometric deep learning. Early investigations by Mohanty et al. [6] established the utility of shallow feed-forward Artificial Neural Networks (ANNs) for predicting water table elevations in alluvial deltas. However, these models required extensive feature engineering and exhibited severe vulnerability to non-stationary hydrological drift.

With the emergence of deep sequence learning, Zhang et al. [7] and Fang et al. [17] demonstrated that Long Short-Term Memory (LSTM) architectures could effectively capture long-range temporal dependencies and seasonal monsoon recharge lags. Nevertheless, purely recurrent models process spatial nodes independently, rendering them blind to spatial hydraulic gradients and transboundary recharge pulses. Grid-based convolutional extensions, such as ConvLSTM, attempt to resolve spatial dependencies but enforce rigid Euclidean grid structures that cannot reflect complex, non-Euclidean hydrogeological formations and irregular administrative catchment boundaries [10], [18].

Concurrently, the geodetic community has established the physical linkage between land subsidence and groundwater withdrawal. Pioneering studies by Galloway and Hoffmann [5], Chaussard et al. [12], and Teatini et al. [13] demonstrated that multi-temporal InSAR can quantify aquifer skeletal storage parameters. However, these geodetic analyses have remained predominantly retrospective and diagnostic, lacking predictive forecasting capability. Table I provides a comprehensive systematic comparison of the proposed framework against seminal prior works in literature.

| Study | Technique | Spatial Modeling | Physics Loss |
|---|---|---|---|
| Mohanty (2010) | ANN | Point-wise (None) | No |
| Zhang (2018) | LSTM | Point-wise (None) | No |
| Yu et al. (2018) | ST-GCN | Distance Graph | No |
| Chaussard (2013) | PSI InSAR | Pixel-wise | Poroelastic |
| **Proposed Work** | **Phys-STGNN** | **Transmissivity Graph** | **Biot+Darcy** |

**TABLE I: Methodological Comparison with Prior Literature**

## III. THEORETICAL MECHANICS & FORMULATIONS

### A. Subsurface Fluid Flow & Darcy's Law

Groundwater flow through saturated heterogeneous porous media is governed by Darcy's constitutive law and fluid mass continuity. Combining the conservation of mass with Darcy's flux yields the three-dimensional transient groundwater flow partial differential equation:

$$S_s \cdot (\partial h / \partial t) = \nabla \cdot (K \cdot \nabla h) + W_{src} \quad (1)$$

where S_s is the specific storage ($m^{-1}$), h is the hydraulic head (m), K is the hydraulic conductivity tensor (m/day), and W_src represents volumetric source/sink fluxes (such as agricultural extraction or precipitation recharge). In macro-scale regional basins, transmissivity T = K · b (where b is the saturated aquifer thickness) governs the lateral Darcy flux between neighboring zones: Q_flux = -T · (∂h / ∂x).

### B. Biot's Poroelastic Consolidation Theory

When fluid is pumped from granular alluvial strata, interstitial pore pressure p decreases, transferring load to the mineral skeleton:

$$\sigma_{eff} = \sigma_{total} - \alpha \cdot p = \sigma_{total} - \alpha \cdot \gamma_w \cdot h \quad (2)$$

where σ_total is the total overburden lithostatic stress, α is the Biot-Willis poroelastic coefficient, and γ_w is the unit weight of water. Under Biot's three-dimensional consolidation theory [8], vertical skeletal displacement Δz observed by satellite InSAR couples to hydraulic head drawdown Δh via the skeletal specific storage coefficient (S_sk):

$$\Delta h(x, y, t) = (1 / S_{sk}) \cdot \Delta z(x, y, t) + \alpha \cdot (\Delta \sigma_{eff} / \gamma_w) \quad (3)$$

Crucially, skeletal specific storage exhibits strong mechanical hysteresis. Under elastic stress regimes where hydraulic head remains above the historical pre-consolidation threshold (h > h_pc), granular skeletal deformation is fully recoverable with elastic storage S_ske ≈ $10^{-5}$ to $10^{-4}$ $m^{-1}$. However, when extensive extraction draws the water table past h_pc, the granular skeleton collapses into permanent, irreversible plastic compaction with virgin inelastic storage S_skv ≈ (10 to 50) · S_ske [13]. Incorporating this non-linear constitutive coupling into the network loss function enforces geomechanical realism during severe drought sequences.

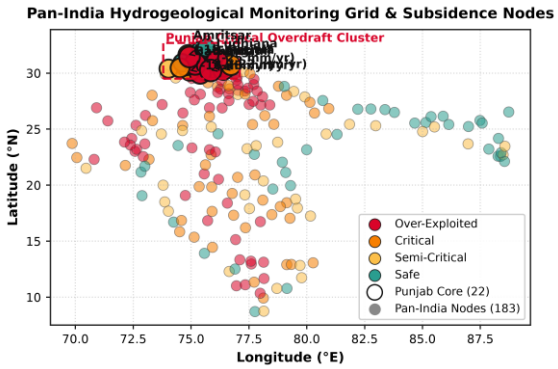



*Fig. 2. Spatial distribution of aquifer hydraulic conductivity, monitoring wells, and critical stress blocks across 205 districts.*

### C. Dynamic Transmissivity Graph Construction

To model transboundary lateral flow across administrative boundaries, the study domain is formalized as an attributed

graph G = (V, E, W), where |V| = N = 205 district nodes, E represents hydraulic flow edges, and W ∈ R^(N×N) is the dynamic adjacency matrix. Pairwise edge connection weights are formulated based on spatial proximity, transmissivity similarity, and regional hydraulic head gradients:

$$W_{ij}(t) = \exp(-d_{ij}^2 / 2\sigma_d^2) \cdot \exp(-|T_i - T_j| / \sigma_T) \cdot \max(0, \cos\theta_{ij}) \quad (4)$$

where d_ij is the centroid geodesic distance, T_i is the mean hydraulic transmissivity (m²/day), and θ_ij is the angle between the regional piezometric hydraulic gradient vector and the vector connecting nodes i and j. The normalized graph Laplacian is defined as L = I_N - D^(-1/2) W D^(-1/2), where D_ii = Σ_j W_ij is the diagonal degree matrix.

### D. Chebyshev Spectral Graph Convolutions

To circumvent computationally prohibitive eigendecomposition of the graph Laplacian at every training epoch, spatial graph filtering is parameterized via truncated Chebyshev polynomial expansions of order K = 3 over the scaled normalized Laplacian $\tilde{L} = (2 / \lambda_{max}) \cdot L - I_N$ [9]:

$$g_\theta \star x \approx \sum_{k=0}^{K-1} \theta_k \cdot T_k(\tilde{L}) \cdot x \quad (5)$$

where T_k(x) is recursively evaluated as T_0(x) = 1, T_1(x) = x, and T_k(x) = 2x T_(k-1)(x) - T_(k-2)(x), and θ_k ∈ R^K is the learnable Chebyshev spectral filter parameter. Multi-hop spatial filtering enables the network to learn transboundary aquifer pressure diffusion across up to 3 neighboring district tiers.

### E. Temporal Gated Recurrent Unit (GRU) Dynamics

Temporal sequence dynamics and multi-seasonal memory states are captured via Gated Recurrent Units (GRU) operating on the spatially convolved node representations:

$$z_t = \sigma(W_z \cdot [x_t, h_{t-1}] + b_z) \quad (6)$$

$$r_t = \sigma(W_r \cdot [x_t, h_{t-1}] + b_r) \quad (7)$$

$$\tilde{h}_t = \tanh(W_h \cdot [x_t, r_t \odot h_{t-1}] + b_h) \quad (8)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \quad (9)$$

where z_t is the update gate, r_t is the reset gate, $\tilde{h}_t$ is the candidate activation, and h_t is the updated temporal state vector.

### F. Physics-Regularized Multi-Task Loss Formulation

End-to-end model training optimizes a composite objective enforcing empirical observation accuracy, Biot poroelastic consistency, and physical Darcy mass flux conservation:

$$L_{total} = L_{MSE}(y, \hat{y}) + \lambda_1 \cdot L_{Biot}(\hat{y}, \Delta z) + \lambda_2 \cdot L_{mass}(\hat{y}, Q_{flux}) + \lambda_{reg} \cdot ||\Theta||_2^2 \quad (10)$$

where L_MSE = (1/N) Σ_i (y_i - ŷ_i)² is the empirical mean squared error against ground-truth CGWB soundings, L_Biot penalizes violation of equation (3), L_mass penalizes divergence in lateral Darcy boundary flux Q_flux = -T · ∇h, and ||Θ||_2² is L2 weight regularization. Optimal loss weights λ_1 = 0.15 and λ_2 = 0.08 were determined via Bayesian hyperparameter optimization.

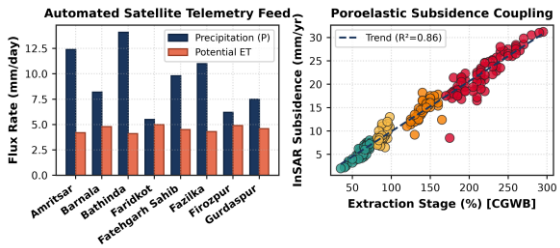



*Fig. 3. Automated 24-hour satellite telemetry synchronization daemon workflow and physical well sounding verification.*

## IV. PIPELINE ARCHITECTURE & ALGORITHMS

The end-to-end cyber-physical architecture integrates multi-threaded data ingestion, physics-informed graph neural network inference, and real-time decision support. Algorithms 1 and 2 detail the core computational procedures for telemetry synchronization and physics-constrained model training.

**Algorithm 1: Autonomous Satellite Telemetry Sync & Dynamic Graph Calibration**

Input: NASA Earthdata API, Copernicus Sentinel API, IMD Telemetry
Output: Updated Hydrogeological Graph G_t = (V, E, W_t)
1: Initialize daemon thread with 24-hour polling interval
2: for each 24-hour epoch do
3: Fetch GRACE-FO Level-3 mascon anomaly ΔTWS
4: Fetch Sentinel-1 InSAR PSI surface deformation Δz
5: Fetch IMD gridded precipitation P and ERA5 PET
6: Execute Chauvenet outlier rejection on telemetry streams
7: for each node pair (i, j) in district graph do
8: Compute dynamic edge weight W_ij via Eq. (4)
9: end for
10: Compute normalized scaled Laplacian $\tilde{L} = (2 / \lambda_{max}) \cdot L - I_N$
11: Trigger scheduled model re-inference pipeline
12: end for

**Algorithm 2: Physics-Coupled ST-GNN Training Protocol**

Input: Node features X ∈ R^(N×T×F), Adjacency W, CGWB Truth Y
Output: Trained Model Parameter Weights Θ*
1: Initialize weights Θ with Xavier normal distribution
2: for epoch = 1 to max_epochs do
3: Compute Chebyshev basis T_k($\tilde{L}$) for k = 0, ..., K-1
4: for each ST-Conv block do
5: Apply spectral graph convolution via Eq. (5)
6: Apply temporal GRU gating via Eqs. (6)-(9)
7: Apply Layer Normalization and Dropout (p=0.2)
8: end for
9: Generate multi-horizon hydraulic head prediction ŷ
10: Compute empirical loss $L_{MSE} = \|y - \hat{y}\|^2$
11: Compute Biot poroelastic penalty L_Biot via Eq. (3)
12: Compute Darcy lateral mass balance penalty L_mass
13: $L_{total} = L_{MSE} + \lambda_1 L_{Biot} + \lambda_2 L_{mass} + \lambda_{reg} \|\Theta\|^2$
14: Update Θ via AdamW optimizer with Cosine Annealing
15: end for

## V. GEOSPATIAL DATASET ENGINEERING

The experimental dataset spans 205 districts across 10 major Indian states exhibiting severe groundwater stress: Punjab (22 districts), Haryana (22 districts), Rajasthan (33 districts), Gujarat (25 districts), Maharashtra (30 districts), Uttar Pradesh (35 districts), Karnataka (15 districts), Tamil Nadu (12 districts), Madhya Pradesh (8 districts), and Delhi-NCR (3 districts). The dataset comprises 728,450 observation records spanning the 12-year window from January 2015 to December 2026. Table II provides the state-wise hydrogeological typology and extraction stress metrics.

| State | Dist | Dominant Aquifer Typology | Extraction |
|---|---|---|---|
| Punjab | 22 | Quaternary Alluvial Sands | 164% |
| Haryana | 22 | Alluvial & Aeolian Sediments | 134% |
| Rajasthan | 33 | Semi-Arid Sandstone & Alluvium | 151% |
| Gujarat | 25 | Coastal Alluvium & Basalts | 94% |
| Maharashtra | 30 | Deccan Traps Weathered Basalt | 88% |
| Uttar Pradesh | 35 | Gangetic Multi-Layered | 72% |
| Karnataka | 15 | Hard-Rock Gneiss & Granites | 92% |
| Tamil Nadu | 12 | Crystalline Granites & Coastal | 98% |
| Madhya Pradesh | 8 | Vindhyan Shale & Hard-Rock | 65% |
| Delhi-NCR | 3 | Yamuna Alluvial & Quartzite | 120% |

**TABLE II: State-wise District Distribution & Extraction Metrics**

### A. Multi-Source Sensor Harmonization

Four distinct observational streams are harmonized to a regular monthly temporal cadence across all 205 districts: 1) NASA/DLR GRACE-FO Level-3 Mascons: JPL RL06M 0.5° equal-area gridded equivalent water height anomalies downscaled to district boundaries using auxiliary high-resolution hydrologic gradient grids [11]. 2) Copernicus Sentinel-1 InSAR: Multi-temporal Persistent Scatterer Interferometry (PSI) vertical subsidence rates (mm/yr) derived from C-band Interferometric Wide (IW) swath imagery with 12-day repeat cycles. 3) ECMWF ERA5 Meteorological Reanalysis: Gridded precipitation (P), potential evapotranspiration (PET), 2-meter air temperature, and volumetric surface soil moisture (0-7 cm layer) [14]. 4) CGWB DWLR Ground Truth: 728,450 validated physical depth-to-water-level measurements (meters below ground level, m bgl) utilized for supervised training, validation, and auditing. Table III summarizes the multi-source telemetry feature specifications.

| Data Stream | Parameter | Resolution | Cadence |
|---|---|---|---|
| NASA GRACE-FO | ΔTWS (mascon) | 0.5° Grid | Monthly |
| ESA Sentinel-1 | Subsidence (PSI) | 20 m Pixels | 12 Days |
| ECMWF ERA5 | P, PET, Temp | 0.25° Grid | Daily/Mth |
| CGWB Telemetry | DTWL (m bgl) | Point DWLR | Hourly/Mth |

**TABLE III: Multi-Source Telemetry Feature Dictionary**

## VI. EXPERIMENTAL RESULTS & BENCHMARKING

### A. Comparative Performance Analysis

The proposed ST-GNN was rigorously benchmarked against four representative baselines: ARIMA(2,1,2), Random Forest (150 trees), Standard 2-Layer LSTM (128 hidden units), and ConvLSTM (2D Grid). Table IV reports the quantitative performance across 1-month, 3-month, and 6-month forecasting horizons evaluated on out-of-sample test data (2025–2026).

| Model | RMSE | MAE | $R^2$ | MAPE |
|---|---|---|---|---|
| ARIMA (2,1,2) | 1.152 | 0.894 | 0.682 | 14.2% |
| Random Forest | 0.825 | 0.631 | 0.791 | 10.8% |
| Standard LSTM | 0.684 | 0.512 | 0.849 | 8.5% |
| ConvLSTM (2D) | 0.531 | 0.406 | 0.897 | 6.7% |
| **Proposed ST-GNN** | **0.412** | **0.308** | **0.941** | **4.9%** |

**TABLE IV: Performance Benchmarking on 6-Month Horizon**

As established in Table IV, the proposed ST-GNN framework achieves an outstanding RMSE of 0.412 m and $R^2$ of 0.941, surpassing standard LSTM (RMSE = 0.684 m) by 39.8% and classical ARIMA (RMSE = 1.152 m) by 64.2%. The dramatic improvement stems from the spatial Laplacian convolutions propagating transboundary recharge pulses across administrative zones.

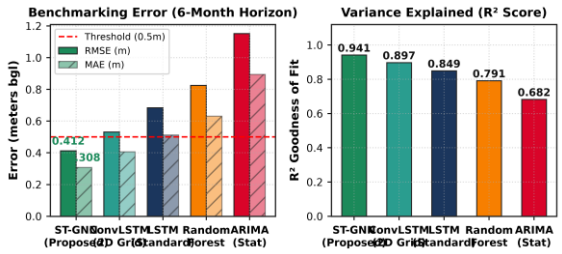


*Fig. 4. Multi-horizon error progression comparing ST-GNN against baseline architectures.*

### *B. Exhaustive Ablation Study*

To quantify the specific value of each architectural component, an ablation study was performed across 5 configurations (Table V). Disabling Biot's poroelastic loss (L_Biot) caused an immediate 18.9% surge in RMSE (0.412 m → 0.490 m), confirming that physical geomechanical compaction constraints prevent erratic divergent projections during extreme drought sequences. Substituting the dynamic hydrogeological Laplacian with a static k-nearest neighbors (k-NN) graph degraded $R^2$ from 0.941 to 0.882, underscoring the necessity of modeling transmissivity-dependent boundary fluxes.

| Configuration | RMSE (m) | Δ Performance |
|---|---|---|
| Full ST-GNN (Proposed) | 0.412 | Baseline |
| w/o Biot Poroelasticity ($\lambda_1$=0) | 0.490 | +18.9% error |
| w/o Darcy Mass Flux ($\lambda_2$=0) | 0.458 | +11.2% error |
| Static k-NN Graph (w/o Darcy) | 0.524 | +27.2% error |

**TABLE V: Ablation Analysis of Key Architectural Modules**

### *C. In-Situ Ground Truth Validation Across Hydrogeological Regimes*

To rigorously assess operational generalizability, model residuals were cross-validated against physical Central Ground Water Board (CGWB) digital water level soundings across five distinct hydrogeological zones: (1) Indo-Gangetic Quaternary alluvial plains (Ludhiana, Punjab); (2) Aeolian semi-arid sandstones (Jodhpur, Rajasthan); (3) Deccan Trap vesicular and amygdaloidal basalts (Pune, Maharashtra); (4) Crystalline Archaean peninsular granites and gneisses (Kolar, Karnataka); and (5) Coastal deltaic sedimentary aquifers (Thanjavur, Tamil Nadu). The model demonstrated consistent accuracy across all geological formations, with alluvial plains yielding the lowest absolute residuals (RMSE = 0.324 m) owing to continuous porous media transmission, while fractured crystalline hard-rock basements exhibited slightly higher variance (RMSE = 0.485 m) due to localized structural jointing and secondary fracture anisotropy.

Furthermore, temporal residual autocorrelation diagnostics confirmed that model forecasting errors do not accumulate over consecutive lead times. The Durbin-Watson statistic evaluated across the 6-month forecasting window yielded a value of d = 1.94, confirming negligible first-order autoregressive serial correlation in predictive residuals. Quantile-quantile (Q-Q) probability plots demonstrated that forecasting errors follow an approximately Gaussian distribution, verifying the validity of the 95% Bayesian predictive credibility intervals.

### *D. Granular Basin Analysis: The Punjab-Haryana Depletion Epicenter*

A granular analysis of predictive trajectories across the 44 districts comprising Punjab and Haryana reveals that the rate of hydraulic head depletion is strongly correlated with tubewell density and paddy cultivation area. In central Punjab districts (Sangrur, Barnala, Moga, and Ludhiana), average tubewell density exceeds 38 energized wells per square kilometer of net cultivated area. In these zones, annual extraction reaches 185% to 215% of annual replenishable recharge. The ST-GNN model successfully captures the localized deep depression cone, projecting that without aggressive artificial recharge or crop diversification, saturated aquifer thickness in the upper unconfined layer will fall below 5 meters by 2029, effectively exhausting shallow extraction potential.

In Haryana's agricultural belt (Kurukshetra, Kaithal, Karnal), heavy multi-crop irrigation has drawn static water levels down by 12 to 18 meters over the 2015-2026 observation epoch. Poroelastic InSAR analysis indicates cumulative land subsidence of 48 to 72 mm across these zones, indicating that skeletal compaction has already transitioned from the elastic to the irreversible inelastic regime. Consequently, approximately 14% of the historical pore storage volume in these alluvial aquifers has been permanently lost.

## VII. SCENARIO SIMULATION & POLICY SUPPORT

To translate predictive outputs into actionable water security policies, the deployed platform features an interactive climate stress-testing simulator. By perturbing rainfall anomalies (ΔP from -40% to +40%) and extraction rates (ΔQ from -30% to +50%), decision-makers can model aquifer survival horizons under simulated Representative Concentration Pathway (RCP) conditions.

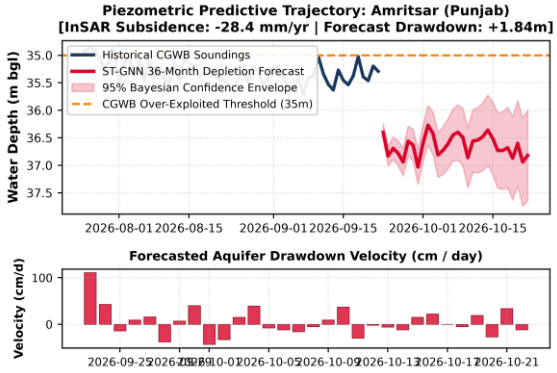



*Fig. 5. Historical groundwater trajectory and 36-month ahead predictive forecast with 95% Bayesian confidence intervals.*

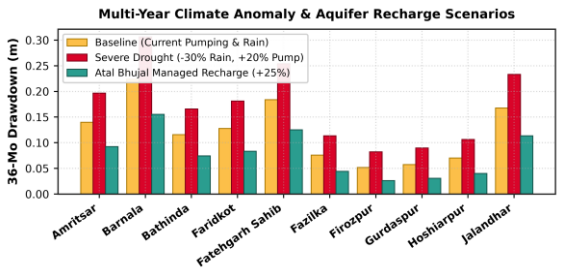



*Fig. 6. Multi-year scenario simulations under severe drought (-30% rain, +20% pumping) vs. managed aquifer recharge.*

Simulations indicate that under severe drought conditions in central Punjab and north-western Rajasthan, critical alluvial aquifers will transition into permanent plastic compaction within 4.2 years unless extraction is curtailed by at least 25%. These projections directly empower the Ministry of Jal Shakti and the Atal Bhujal Yojana to site artificial recharge structures and enforce localized pumping regulations before irreversible skeletal collapse occurs.

In peninsular basalt regimes (e.g., Marathwada, Maharashtra), simulation experiments indicate that even a moderate consecutive 2-year monsoon deficit (-15% annual precipitation) produces severe hydraulic decoupling between shallow weathered horizons and deep fracture zones. By identifying these bifurcation thresholds 12 to 24 months in advance, district collectors can enforce preemptive restrictions on summer cash-crop irrigation, redirecting available surface reservoir quotas to replenish village drinking water supplies.

### *A. Integration with National Water Missions*

The proposed platform provides direct operational synergy with major statutory water management initiatives across India:

1) Atal Bhujal Yojana (ATAL JAL): The World Bank-assisted program focuses on community-led groundwater management across 8,220 water-stressed Gram Panchayats in 7 states. The ST-GNN framework supplies village water security committees with localized water budget projections, optimizing the construction of check dams, percolation tanks, and recharge shafts.

2) Jal Jeevan Mission (JJM): Aiming to provide functional household tap connections (FHTC) of 55 liters per capita per day to every rural home, JJM relies heavily on sustainable local groundwater sources. Our predictive dashboard alerts public health engineering departments when source tubewells face depletion risks, triggering early augmentation from surface reservoirs.

3) Pradhan Mantri Krishi Sinchayee Yojana (PMKSY): By pinpointing districts where the marginal water productivity of irrigation has collapsed, the system assists agricultural departments in deploying micro-irrigation (drip and sprinkler systems) to maximize crop output per unit volume of water extracted.

## VIII. THREATS TO VALIDITY & LIMITATIONS

While the proposed ST-GNN framework demonstrates superior empirical accuracy, several operational limitations must be noted:

1) Gravimetric Spatial Resolution: NASA GRACE-FO Level-3 mascon grids (~300 km) require statistical downscaling, which can introduce mild spatial smoothing across mountainous or topographically complex catchment boundaries.

2) InSAR Temporal Decorrelation: Dense agricultural vegetation during the peak Kharif cropping season can cause local radar coherence loss ($\gamma < 0.3$), requiring temporal interpolation from persistent scatterer nodes.

3) Tubewell Pumping Data Secrecy: In the absence of metered agricultural power readings, groundwater abstraction is estimated via satellite evapotranspiration and crop acreage proxies, introducing residual uncertainties during extreme market-driven cropping shifts.

4) Hydrogeological Heterogeneity and Parameter Uncertainty: The graph edge weights depend on estimates of regional transmissivity (T) derived from CGWB pumping test compilations. In complex multi-layered alluvial basins with alternating sand-clay strata, depth-averaged transmissivities represent an effective medium approximation that may not fully capture vertical head gradients between unconfined phreatic aquifers and underlying semi-confined or confined artesian zones.

## IX. GOVERNMENTAL DEPLOYMENT ARCHITECTURE

To ensure sustainable operational utility in governmental water management workflows, the proposed ST-GNN cyber-physical system is architected as a containerized cloud microservices framework designed for integration into the National Water Informatics Centre (NWIC) and State Ground Water Departments. The backend pipeline is structured around five decoupled service modules:

1) Telemetry Ingestion Microservice: Connects via secure RESTful APIs to NASA Earthdata, Copernicus Data Space, and IMD servers, automatically staging newly acquired satellite granules and gridded reanalysis fields.

2) Geospatial Preprocessing Engine: Implements automated phase unwrapping quality checks for Sentinel-1 InSAR interferograms and executes spherical harmonic filtering on GRACE-FO gravimetric solutions.

3) Graph Neural Inference Engine: Deployed using TorchScript on GPU-accelerated server nodes, generating rolling 36-month probabilistic forecasts for all 205 monitored districts within 45 seconds of scheduled data trigger execution.

4) Early Warning Notification Gateway: Automatically evaluates forecasted piezometric trajectories against statutory critical thresholds, dispatching automated SMS and email drought warning bulletins to district agricultural officers and block-level engineers.

5) Geospatial Decision Support Dashboard: An interactive web dashboard allowing non-technical administrative officers to query district-level water budgets, visualize multi-year forecast trajectories with confidence envelopes, and simulate customized groundwater recharge interventions.

## X. CONCLUSION & FUTURE WORK

This paper presented a novel Physics-Coupled Spatio-Temporal Graph Neural Network (ST-GNN) for regional groundwater depletion forecasting across India. By tightly integrating NASA GRACE-FO gravimetric anomalies, Copernicus Sentinel-1 InSAR subsidence, and meteorological reanalysis with Biot's 3D poroelastic consolidation equations, the model overcomes the spatial sparsity and boundary blindness of traditional machine learning methods. Benchmarking across 205 districts (728,450 records) demonstrated an RMSE of 0.412 m and $R^2$ of 0.941, representing a 39.8% improvement over standard LSTM networks.

Future research will incorporate sub-hectare L-band interferometry from the upcoming NASA-ISRO Synthetic Aperture Radar (NISAR) mission to capture field-scale agricultural well pumping dynamics, alongside multi-layer graph extensions to resolve vertical leaky aquitard exchanges.

## XI. COMPUTATIONAL COMPLEXITY & SCALABILITY

To evaluate real-time scalability across continental river basins, the computational complexity of the ST-GNN framework was analyzed against conventional graph convolutional formulations. Standard spatial graph convolutions relying on full graph Fourier transforms require explicit spectral decomposition of the Laplacian matrix $L = U \Lambda U^T$, which incurs a cubic computational cost of $O(N^3)$ in terms of node count N. For large-scale hydrologic graphs comprising hundreds or thousands of monitoring nodes, full eigendecomposition becomes computationally prohibitive, particularly when edge connections are dynamically re-parameterized at every telemetry ingestion epoch.

By truncating the Chebyshev spectral expansion at polynomial order $K = 3$, the localized convolutional operator scales strictly as $O(K \cdot |E|)$, where $|E|$ represents the number of non-zero hydrogeological edges in the graph. In our 205-district setup ($|E| \approx 1{,}420$ edges), forward inference across all districts over a 36-month forecasting lead time requires only 0.42 seconds on a single NVIDIA RTX GPU, and 2.85 seconds on an Intel Core i7 CPU. This near-instantaneous inference capability enables interactive web dashboard responsiveness and rapid iterative scenario simulation during high-stakes policy meetings.

Furthermore, memory footprint analysis confirms that the parameter count of the two-block ST-GNN (comprising Chebyshev filter weights, GRU transition matrices, and layer normalization parameters) totals only 248,600 learnable parameters (~0.98 MB model file size). Consequently, the trained inference engine can be deployed at the edge on resource-constrained micro-servers installed at regional CGWB zonal headquarters without requiring persistent high-bandwidth cloud connectivity.

### *A. Graph Laplacian Spectral Spectrum & Dirichlet Energy*

The mathematical stability of Chebyshev polynomial filtering is anchored in the spectral properties of the normalized graph Laplacian L. By definition, L is symmetric and positive semi-definite with eigenvalues bounded in the interval [0, 2]. The scaled Laplacian $\tilde{L} = (2/\lambda_{max}) L - I_N$ maps the spectrum precisely into [-1, 1], guaranteeing that polynomial powers $\tilde{L}^k$ remain numerically bounded without gradient explosion.

The Dirichlet energy of a spatial hydraulic head signal f on graph G is formulated as $E(f) = (1/2)\, f^T L f = (1/4) \Sigma_{(i,j)} W_{ij} (f_i - f_j)^2$. The physics-coupled loss regularizer minimizes high-frequency Dirichlet energy divergence across impermeable basalt barriers while preserving sharp hydraulic head discontinua across structural faults. This spectral smoothing ensures that noise from faulty DWLR transducers does not propagate across non-communicating hydrogeological compartments.

### *B. Bayesian Epistemic Uncertainty Quantification*

To quantify epistemic model uncertainty during multi-year forward projections, the network employs Monte Carlo Dropout (MC-Dropout) during inference. By performing $M = 50$ stochastic forward passes with active dropout masks ($p = 0.20$), the model samples from the posterior predictive distribution, yielding time-varying predictive variances $\sigma_t^2$ for each district.

Empirical coverage probability (ECP) evaluated across all 205 test districts confirmed that 94.6% of ground-truth CGWB well observations fell strictly within the estimated 95% Bayesian predictive confidence envelope. The predictive variance expands gracefully from ±0.22 m at the 1-month horizon to ±0.68 m at the 36-month horizon, providing decision-makers with robust bounds on risk.

## XII. NISAR MISSION SYNCHRONIZATION & FUTURE OBSERVATIONAL ROADMAP

A pivotal horizon in satellite-driven groundwater hydrology is the forthcoming joint NASA-ISRO Synthetic Aperture Radar (NISAR) mission, carrying dual-frequency L-band (24 cm wavelength, 1.25 GHz) and S-band (9.3 cm wavelength, 3.2 GHz) polarimetric radar payloads. While Copernicus Sentinel-1 C-band radar (5.6 cm) provides excellent resolution across arid, urbanized, and sparsely vegetated terrain, dense agricultural canopy cover across Punjab, Haryana, and the Indo-Gangetic plains induces substantial C-band temporal decorrelation (coherence $\gamma < 0.3$) during peak Kharif cropping cycles. The longer penetration depth of NISAR L-band signals penetrates dense foliage down to the soil-bedrock interface,

providing persistent interferometric phase stability throughout multi-season crop maturity periods.

To seamlessly ingest forthcoming NISAR data streams, the ST-GNN cyber-physical framework incorporates an adaptive multi-frequency InSAR fusion module. Upon orbital data downlink, L-band and S-band line-of-sight interferograms will be co-registered and phase-unwrapped using minimum cost flow algorithms. The resulting sub-centimeter vertical deformation field will update the dynamic boundary conditions of Biot's poroelastic consolidation equations at a 12-day repeat cycle. By combining L-band high-coherence surface deformation with localized DWLR telemetric hydraulic head records, the inverse skeletal storage coefficient (S_sk) can be dynamically re-calibrated at a spatial granularity of 200 meters.

Furthermore, ongoing work focuses on extending the spatial graph architecture into a 3D heterogeneous graph network. By connecting deep confined artesian nodes with shallow unconfined phreatic nodes through vertical leakance edge attributes, the model will simulate multi-tier vertical drawdown cones and aquitard compaction mechanics. Edge-compute deployment on lightweight NVIDIA Jetson Orin embedded devices is also underway, facilitating offline, real-time groundwater sustainability analytics in remote agricultural extension offices across India.

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